Web Analytics Final Project Plan

Project Title: Tech Skill Demand & Salary Analysis

Course Requirement Focus: Data Collection via API, Data Analysis, and Interactive Visualizations

1. Project Overview & Research Question

Objective

To analyze real-time tech job market listings to identify which technical skills and work arrangements (Remote vs. On-site) are most demanded by employers and carry the highest salary premiums.

Primary Research Question

"Which specific technical skills (e.g., Python, SQL, Cloud) and work models (Remote vs. Hybrid) command the highest salary premiums in modern tech and data roles?"

Key Sub-Questions

1. What are the top 10 most requested technical skills across job postings?
2. Is there a statistically significant salary difference between fully remote and location-bound positions?
3. Which specialized tools (e.g., AWS, Snowflake, Spark) add the largest dollar premium to a baseline salary?

2. Simple Tool Stack (No Overcomplicated Setup)

To keep execution straightforward, this project relies strictly on accessible, beginner-friendly tools:

Component	**Selected Tool**	**Why It Was Chosen**
Language	Python (pandas, requests)	Standard for data cleaning and API retrieval.
Data Source	**Remotive API** or **Adzuna API**	Free, well-documented REST APIs with no complex scraping logic.
Environment	Google Colab or Jupyter Notebook	Simple, free web-based python environment; no local installation needed.

Component	Selected Tool	Why It Was Chosen
Dashboard	**Tableau Public** or **Power BI**	Free, drag-and-drop dashboard builders (no code required for UI).

3. Data Collection Methodology (API Pipeline)

Data will be retrieved from the **Remotive Public API** or **Adzuna API**, which return structured JSON data containing job title, location, category, salary, and description text.

Implementation Logic:

1. Issue a standard HTTP request to the API endpoint.
2. Parse the returned JSON response into a Pandas DataFrame.
3. Save the raw dataset to CSV for backup.

4. Data Cleaning & Analysis Techniques

Step A: Data Cleaning & Standardizing

- Filter out listings without valid salary data (when analyzing compensation).
- Standardize salary ranges into a single average value:

$$\text{Salary Average} = \frac{\text{Min Salary} + \text{Max Salary}}{2}$$

Step B: Keyword Extraction (Simple Text Matching)

Instead of complex Natural Language Processing, use simple text matching on the job description column to identify required skills:

Python

```
# List of target skills to track
skills = ["python", "sql", "aws", "tableau", "spark", "snowflake", "docker", "r"]


# Check presence of each skill (1 if present, 0 if absent)
for skill in skills:
    jobs_df[skill] = jobs_df["description"].str.lower().str.contains(skill).astype(int)
```

Step C: Analytics & Insights

1. **Demand Percentage:** Calculate the percentage of job postings asking for each skill:

$$\text{Demand Ratio} = \frac{\text{Postings requiring Skill X}}{\text{Total Postings}} \times 100$$

2. **Salary Premium Analysis:** Group by skill columns to calculate median salary for jobs that require the skill versus those that do not.

5. Dashboard Design & Layout Strategy

Once analysis is complete in Python, export the final dataset as job_market_analyzed.csv and load it into **Tableau Public** or **Power BI** to build a simple interactive dashboard.

Dashboard Wireframe / Visual Elements

- **Top Section — KPI Cards:**
    - **Total Jobs Analyzed** (e.g., 850)
    - **Average Market Salary** (e.g., $115,000)
    - **Remote Job Share** (e.g., 62%)
- **Middle Section — Charts:**
    - **Chart 1 (Horizontal Bar Chart):** *Most Requested Skills* (Skill Name vs. % of Job Postings).
    - **Chart 2 (Box Plot or Bar Chart):** *Median Salary by Required Skill* (Comparing base salary vs. skill-boosted salary).
- **Bottom Section — Interactivity:**
    - **Interactive Filter Sidebar:** Dropdowns allowing dashboard viewers to filter all visuals by *Job Category* or *Location*.
    - **Detailed Job List Table:** Displaying Job Title, Company, Salary, and direct URL.

6. Execution Roadmap & Timeline

1. **Phase 1 (Data Retrieval):** Run the Python API script to generate job_market_raw.csv.
2. **Phase 2 (Analytics):** Execute skill extraction logic and salary calculations in Python; export job_market_analyzed.csv.
3. **Phase 3 (Dashboard & Presentation):** Import CSV into Tableau/Power BI, build visuals, and insert screenshots into the final project report.